The rank of a set is the strict supremum of the ranks of its members

The Open Logic Project contributors

openlogic:set-theory/spine:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-spine/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: both inequality paragraphs and the closing equality line are verbatim in the span.

Editorial note. The label ranksupstrict names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Members have strictly smaller rank. Same unit, \ollabel{rankmemberslower} (source line 602). Line 644: one inequality, alpha <= rank(A), from all member-ranks living inside rank(A).
- Hierarchy as the sets of smaller rank. Same unit, \ollabel{valphalowerrank} (source line 589). Line 646: the converse, each x in A landing in V_alpha, hence A subset V_alpha.
- Strict supremum of ordinals. Unit ordinals, definition defsupstrict. The shape of both sides of the claimed equality.

Source reference. Original reference rankmemberslower: sources/olp-set-theory-spine/pinned-original-source.tex, source lines 602–602. Prerequisite pointer invoked at proof line(s) 645; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference valphalowerrank: sources/olp-set-theory-spine/pinned-original-source.tex, source lines 589–589. Prerequisite pointer invoked at proof line(s) 646; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. $\operatorname{rank} A = \subsetneq_{x \in A} \operatorname{rank} x$.

Proof. Let $\alpha = \bigsqcup_{x \in A} \operatorname{rank} x$. By rankmemberslower, $\alpha \leq \operatorname{rank} A$. But if $x \in A$ then $\operatorname{rank} x \in \alpha$, so that $x \in V_\alpha$ by valphalowerrank, and hence $A \subseteq V_\alpha$, i.e., $\operatorname{rank} A \leq \alpha$. Hence $\operatorname{rank} A = \alpha$. □